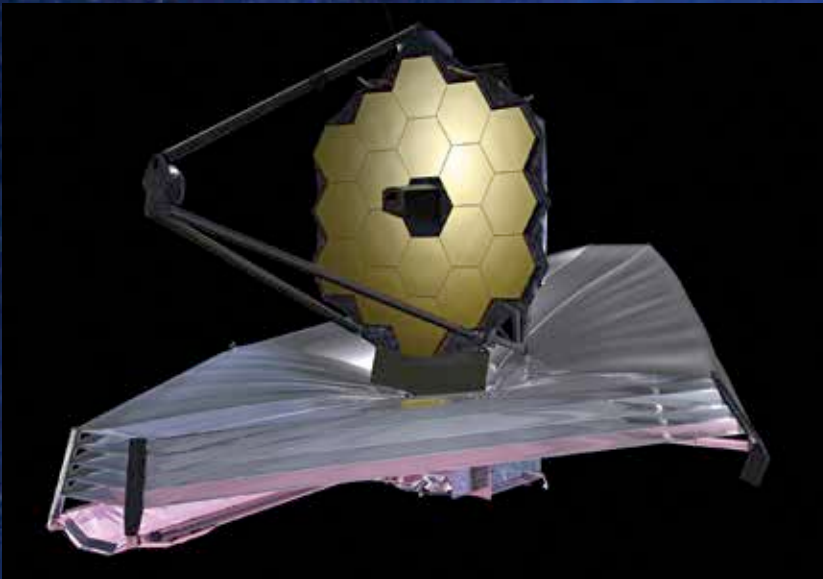




That should not be possible!

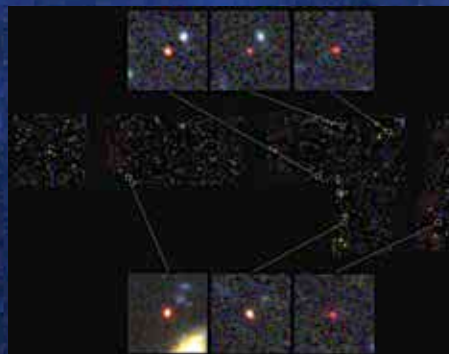
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The James Webb Telescope has become synonymous with space exploration over the last year and a half, at least in popular news and social media. If by any chance you thought it was done for the time being, then you need to reconsider. The telescope that was launched in December 2021 recently sent back data, including images, which, when analysed by a team of international astrophysicists, has revealed a total of six galaxies.

If they were just regular galaxies that had been theorised to exist by researchers at any given point in time, the discovery would not have been regarded as groundbreaking. However, these galaxies that were found by the telescope were so massive that as per the

current cosmological theories, they should not exist at all.

As per the data that was published by the team of researchers as a part of their research paper titled – A population of red candidate massive galaxies ~600 Myr after the Big Bang – in the journal *Nature*, these galaxies existed 'at the dawn of the universe', which equates to about 500 to 700 million years after the Big Bang. Don't sit down and



start running numbers through your calculator. It was more than 13 billion years ago, and their stellar mass is greater than 10^{10} solar masses, with one galaxy possibly having a total stellar mass of $\sim 10^{11}$ solar masses.

Erica Nelson, who is a co-author of the research paper in focus, very aptly describes this discovery. As reported by the University of Colorado Boulder, she said, "It's bananas... You just don't expect the early universe to be able to organize itself that quickly. These galaxies should not have had time to form."

These findings by researcher Nelson and her colleagues are a result of a special endeavour by the James Webb Telescope named – Cosmic Evolution Early Release Science (CEERS) Survey. As a part of this survey, the James Webb Telescope is tasked with looking deep into the universe, specifically at the Big Dipper, a region of space that was first observed in the 1990s by the legendary Hubble Space Telescope. The telescope uses the onboard Near Infrared Camera (NIRCam) for imaging.

In her quotes published by the university in their release, Nelson even compared the James Webb Telescope to the Hubble Space Telescope. While the rate of discoveries made using the JWST is similar to that of the Hubble, which was foreseen by researchers worldwide, as per Nelson, it is still surprising to see such massive breakthroughs becoming a reality.

While there's still a long way to go for the researchers, the importance of this preliminary data and the images revealed by the telescope has the potential to completely change the way we study and know the world of astronomy.

The research paper can be accessed here: <https://digit.in/masgal>